UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 010 · Problems on maxima and minima for a function of two variables · compiled 15 October 2026

Contents

1. Short notes
2. Keyword index
3. Rapid recall
4. Last 10 minutes

1 Short notes

1 · The method, and the ESA cubic functions

$f_x = 0$, $f_y = 0$ → split into cases → stationary points → r, s, t → $rt - s^2$ and r → values. ESA: $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$: max 112 at $(4, 0)$, min 108 at $(6, 0)$, saddles $(5, \pm 1)$. $x^3 - 12x + y^3 + 3y^2 - 9y$: max 43 at $(-2, -3)$, min -21 at $(2, 1)$, saddles $(2, -3)$, $(-2, 1)$.

2 · Functions with many stationary points

$x^4 + y^4 - x^2 - y^2 + 1$: max 1 at $(0, 0)$, min $\frac{1}{2}$ at $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$, four saddles on the axes. $\sin x + \sin y + \sin(x + y)$: $x = y = \frac{\pi}{3}$, max $\frac{3\sqrt{3}}{2}$.

3 · Applied problems: eliminate one variable first

Eliminate with the condition, then use the two-variable test. $400xyz^2$ on the unit sphere: max 50 at $(\pm\frac{1}{2}, \pm\frac{1}{2})$ (same signs); $kxyz^2$ on radius a: $\frac{ka^4}{8}$. Open box, volume V: $A = xy + \frac{2V}{x} + \frac{2V}{y}$, $x = y = \sqrt[3]{2V}$, $z = \frac{x}{2}$.

2 Keyword index

9 terms; the number in brackets is the topic.

Case split (1). $6y(x-5)=0$: $y=0$ or $x=5$.

Elimination (3). Use the constraint to remove one variable.

Extreme values (1). f at the maxima and minima.

Highest temperature (3). $400xyz^2$ on the unit sphere: 50.

Open box (3). $A=xy+2yz+2zx$; $x=y=\sqrt[3]{2V}$, $z=\frac{x}{2}$.

Saddle point (1). $rt-s^2<0$; no extreme value.

Separable function (2). f_x in x only, f_y in y only.

Stationary points (1). Solutions of $f_x=f_y=0$.

Trigonometric stationary points (2). Subtract the conditions: $\cos x=\cos y$.

3 Rapid recall

1	Stationary points of $x^3+3xy^2-15x^2-15y^2+72x$?	$(4,0)$, $(6,0)$, $(5,\pm1)$.
2	Its maximum and minimum?	Max 112 at $(4,0)$; min 108 at $(6,0)$.
3	Its $rt-s^2$?	$36[(x-5)^2-y^2]$.
4	Stationary points of $x^3-12x+y^3+3y^2-9y$?	$(\pm2,1)$, $(\pm2,-3)$.
5	Its extreme values?	Max 43 at $(-2,-3)$; min -21 at $(2,1)$.
6	$x^4+y^4-x^2-y^2+1$: max? min?	Max 1 at $(0,0)$; min $\frac{1}{2}$ at $(\pm\frac{1}{\sqrt{2}},\pm\frac{1}{\sqrt{2}})$.
7	$\sin x+\sin y+\sin(x+y)$?	Max $\frac{3\sqrt{3}}{2}$ at $(\frac{\pi}{3},\frac{\pi}{3})$.
8	Highest temperature $400xyz^2$ on the unit sphere?	50 at $x=y=\pm\frac{1}{2}$.
9	$kxyz^2$ on a sphere of radius a?	$\frac{ka^4}{8}$.
10	Open box of volume V, least area?	$x=y=\sqrt[3]{2V}$, height $\frac{x}{2}$.
11	Is a cube optimal for an open box?	No; only for a closed box.

Mnemonic (study aid). *"Case, point, table, value."* Split the factored equation into cases, collect the points, tabulate r and $rt-s^2$, then substitute for the values.

4 Last 10 minutes before the exam

Top facts

1. Routine: $f_x = f_y = 0$ → points → r, s, t → $rt - s^2$ and r → values.
2. ESA $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$: max 112 at $(4, 0)$, min 108 at $(6, 0)$, saddles $(5, \pm 1)$.
3. ESA $x^3 - 12x + y^3 + 3y^2 - 9y$: max 43 at $(-2, -3)$, min -21 at $(2, 1)$, saddles $(2, -3)$, $(-2, 1)$.
4. $rt - s^2 < 0$: saddle, whatever the sign of r.
5. Constraint present: eliminate one variable (or use Lagrange, class 011).
6. $400xyz^2$ on the unit sphere: highest temperature 50.
7. Open box: base $\sqrt[3]{2V}$ square, height half the base edge.

Top formulas

E1.1 · Stationary points

ESA ★★★

$$f_x = 0, \qquad f_y = 0 \quad \text{(solve simultaneously)}$$

GIVES All candidate points.

APPLIES Every problem, first step.

NOTE Factorise: $6xy - 30y = 6y(x - 5) = 0$ gives the cases $y = 0$ or $x = 5$.

S010·7 · T010b 00:01:23 · T010b 00:01:40

E1.2 · The test

ESA ★★★

$$r = f_{xx}, \quad s = f_{xy}, \quad t = f_{yy}$$

$$rt - s^2 > 0 :\ r < 0 \text{ max},\ r > 0 \text{ min}; \qquad rt - s^2 < 0 :\ \text{saddle}; \qquad rt - s^2 = 0 :\ \text{doubtful}$$

GIVES The nature of each stationary point.

APPLIES At each point separately.

NOTE Restated from class 009.

S010·8 · T010b 00:03:12 · T010b 00:04:02

E1.3 · Hessian of the ESA cubic

ESA ★★★

$$f = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x : \qquad rt - s^2 = 36\left[(x - 5)^2 - y^2\right]$$

GIVES $rt - s^2$ at every point at once.

APPLIES Dec-23 1c, Jan-24 s1 1a.

NOTE $r = t = 6x - 30$, $s = 6y$. Slide 010·8 drops the 36 on y^2.

S010·8 · T010b 00:03:30

E3.1 · Eliminating with a sphere

ESA ★★

$$x^2+y^2+z^2=a^2 \;\Rightarrow\; z^2=a^2-x^2-y^2, \qquad T=kxyz^2=kxy(a^2-x^2-y^2)$$

GIVES A function of x, y only.

APPLIES Quantity contains z^2 and the point lies on a sphere.

NOTE Unit sphere: $a=1$.

S010·3 · S010·9 · T010a 00:01:00

E3.2 · Open box of volume V

ESA ★★

$$A=xy+2yz+2zx, \quad xyz=V \;\Rightarrow\; A=xy+\frac{2V}{x}+\frac{2V}{y}$$

GIVES Surface area (metal used) of a box with no top, base $x \times y$.

APPLIES Least material for a given volume.

NOTE Minimum at $x=y=\sqrt[3]{2V}$, $z=\frac{x}{2}$ (CW P34). The slide's $A=2xy+2yz+zx$ is not the area of an open box.

ADDED FOR CLARITY S010·5 · CW U1 P34